COLLEGE OF ENGINEERING & TECHNOLOGY				
	Department:	**Basic and Applied Sciences**		
	Course title:	Physics (1)	**Course Code**	EBA1110
	Lecturer	**Dr. Zeinab Mahmoud Ahmed**		
	Textbook:	Physics for Scientist and Engineers with Modern physics		

Sheet (1) Vectors Components, Addition, Subtraction & Multiplication

1) A displacement vector has a magnitude of 12.0 km in the direction 40° west of north. What is the north component of this vector? The west component?

2) A vector length of 5.0 m is in the *x*–*y* plane at an angle of 30° with the *x*-axis. What is the x component of this vector? The y component?

3) Suppose that $A = -5i - 3j + k$ and $B = 2.0i + j - 3.0k$. Calculate the following:

(a) A +B

(b) A -B

(c) 2A -3B

4) A person walks 25.0° north of east for 3.10 km. How far would she have to walk due north and due east to arrive at the same location?

5) A girl delivering newspapers covers her route by traveling 3.00 blocks west, 4.00 blocks north, and then 6.00 blocks east. What is her resultant displacement? What is the total distance she travels?

6) A sailboat tacking against the wind moves as follows: 3.2 km at 45° east of north, 4.5 km at 50° west of north, and 2.6 km at 45° east of north. What is the net displacement for the entire motion?

7) A man runs 2.5 km north and then 1.5 km in a direction 30° east of north. A woman walks directly between the same initial and final points. What distance does the woman walk? In what direction?

8) An airplane flies 480 km in a direction 40° east of north to a hub city, and then 370 km in a direction 10° north of west to a final destination. Determine the magnitude and direction of the net displacement.

9) Calculate the dot product for the vectors $5.0i - 2.0j + 1.0\,k$ and $2.0i - 1.0\,k$

10) For the vectors $A = 4.0i + 3.0j + 2.0k$ and $B = -1.0i + 2.0j + 1.0\,k$. Calculate A.B